

KRISHNA VARUN KASAMNENI

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WORK EXPERIENCE

Platform Software Engineer -1 | Full-time

Jul 2025 - Present
Bengaluru

Oracle

- Engineered and maintained AI-assisted test automation covering **300+ Oracle Blockchain Platform APIs**, strengthening release validation and reducing manual verification effort.
- Built AI-agent and structured-prompt workflows using **Codex and LangChain** to translate requirements into Java/TestNG test designs, generate automation scaffolds, strengthen assertions, analyze failures, and extend reusable testing utilities.
- Developed a reusable end-to-end API testing framework for **Hyperledger Fabric** services, request construction, response validation, transaction-state verification, assertion logic, and diagnostic reporting.
- Automated validation across **seven chaincode asset-lifecycle workflows**—minting, transfers, approvals, holds, burns, balance checks, and transaction-history verification.
- Designed and executed **40+ regression scenarios across 10 platform areas**, including wrapper and console APIs, chaincode lifecycle.

Technologies / Skills Used : Oracle Cloud Infrastructure, Hyperledger Fabric, Lumberjack, Java, TestNG, AI, Codex, Langchain

Data Engineer Intern | Internship

Jun 2024 - Jul 2024
Bengaluru

SIMPL PAY (BNPL)

- Replaced a **once-daily database snapshot job** with a real-time **change data capture (CDC) pipeline**, shifting synchronization from 24-hour batch cycles to continuous event streaming.
- Deployed **Debezium on Kafka Connect** and configured PostgreSQL source connectors to capture row-level inserts, updates, and deletes and publish change events to Kafka topics.
- Integrated Kafka Connect with Kafka brokers to deliver reliable, low-latency database change events, improving data freshness for downstream systems and eliminating reliance on full daily snapshots.

Technologies / Skills Used : Apache Kafka, Kafka Connect, Debezium, PostgreSQL, Change Data Capture (CDC), Real-Time Data Streaming, Event-Driven Architecture, Database Synchronization, Data Pipelines, Distributed Systems

PROJECTS

IssueForge — Multi-Agent GitHub Issue-to-PR Orchestrator

- Engineered an end-to-end multi-agent coding platform that transforms GitHub issues into implementation plans, validated code patches, regression tests, and human-approved pull requests.
- Designed a stateful **seven-stage LangGraph workflow** comprising Code Reader, Planner, Code Writer, Test Writer, sandbox execution, human approval, and PR delivery agents.
- Implemented security and reliability guardrails using strict **Pydantic schemas**, prompt-injection defenses, safe-path validation, deterministic diff generation, bounded retries, and configurable cost budgets.
- Executed AI-generated changes within an offline, least-privilege **Docker sandbox**, enforcing read-only access and CPU, memory, process, network, and privilege restrictions.
- Added **SQLite checkpointing** to persist workflow state and support pausing, inspection, human approval, revision cycles, failure recovery, and resumable execution.
- Integrated GitHub issue ingestion, isolated **Git worktrees**, automated commits, authenticated pull-request creation, execution telemetry, and secret-safe workflow reporting.

Technologies / Tools Used : Python, LangGraph, Pydantic, Docker, Git, GitHub API, SQLite, Pytest, OpenRouter, Multi-Agent Systems, LLM Orchestration, Human-in-the-Loop Workflows

TURBO COMPRESSOR — Lossless Huffman Compression CLI

- Built a C++ command-line utility for lossless file compression and decompression using the **Huffman coding algorithm**.
- Implemented frequency analysis, priority-queue-based Huffman tree construction, prefix-code generation, and bit-level encoding to reduce file size without data loss.
- Developed the corresponding decoding workflow to reconstruct compressed files byte-for-byte in their original form.
- Applied object-oriented design to separate compression, decompression, tree management, and file-handling responsibilities.

Technologies / Tools Used : C++, Huffman Coding, Data Structures and Algorithms, Object-Oriented Programming, CLI Development

SKILLS

Programming Languages : C++, Java, Python, SQL, C

Frameworks & Libraries : LangGraph, LangChain, Spring Boot, Django, Pydantic, TestNG, Pytest

Tools & Platforms : Oracle Cloud Infrastructure (OCI), Hyperledger Fabric, Docker, Kubernetes, Git, GitHub, GitHub API, Bitbucket, Postman, OpenRouter, Lumberjack

EDUCATION

Indian Institute Of Technology (BHU) | Varanasi

Jul 2021 - May 2025
CGPA : 8.46

Bachelor's Degree in Computer Science

Sri Chaitanya Junior College | Vijayawada

Jun 2019 - Jul 2021
Percentage : 96.60

Higher Secondary Education

COMPETITIVE PROGRAMMING

- Expert in **Codeforces**, demonstrating advanced problem-solving skills.
- Achieved a 4 Star ranking on **Codechef**, reflecting consistent performance.
- Earned the title of Knight on **Leetcode**, showcasing expertise in algorithm challenges.

ACADEMIC ACHIEVEMENTS

- Achieved an All India Rank of 1353** in JEE Advanced 2021 among 150,000 candidates.
- Secured an All India Rank of 2435** in JEE Main 2021 among 1 million candidates.
- Recognized as the **Andhra Pradesh State Topper** in the Indian Olympiad in Chemistry (ACT).
- Achieved a perfect 100% score** in Mathematics in JEE Main 2021, demonstrating strong analytical and problem-solving aptitude.